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Course Name: Reinforcement Learning

Course code: MLA0305

Design and Develop Model – Sampling-Based Planning & Data Generation Using Reinforcement Learning

1. AIM

Aim:

To design and develop a sampling-based planning model using **Monte Carlo Tree Search (MCTS)** for intelligent navigation in a grid-world environment, generate synthetic state data, and compare the performance of MCTS with Greedy and Random planning strategies using reward, success rate, number of steps, and collision metrics.

2. PROCEDURE

1. Define an **8 × 8 grid-world environment** with a starting state and a goal state.
2. Generate obstacles randomly in the environment while ensuring that the start and goal positions remain accessible.
3. Generate a synthetic dataset containing **1,000 records and 10 parameters**, including state coordinates, goal coordinates, distance to goal, obstacle density, step cost, goal reward, and collision penalty.
4. Define the available actions of the agent as **Up, Down, Left, and Right**.
5. Assign rewards and penalties:
 - Goal reached → positive reward
 - Normal movement → step cost
 - Collision with obstacle/boundary → penalty
6. Implement **Monte Carlo Tree Search (MCTS)** using four stages:
 - Selection
 - Expansion
 - Simulation/Rollout
 - Backpropagation

7. Run the MCTS agent for multiple episodes to determine suitable actions for reaching the goal.
8. Implement **Greedy** and **Random** agents as baseline models for comparison.
9. Evaluate all three approaches using:
 - Average reward
 - Reward standard deviation
 - Success rate
 - Average number of steps
 - Average collisions
10. Generate comparison graphs showing the performance of the three algorithms.
11. Display the final MCTS navigation path through the generated grid environment.
12. Identify the best-performing algorithm based on the calculated evaluation metrics.

PSEUDO CODE:

BEGIN

Initialize grid, start, goal, actions and rewards

Generate obstacles and 1000-record dataset

FOR each episode:

 Run MCTS:

 Select node

 Expand node

 Simulate actions

 Backpropagate reward

 Select best action

 Move agent and record reward

Run Greedy and Random methods

Calculate reward, success rate,
steps and collisions

Compare all three methods

Generate summary table and graphs

Display final MCTS path

END

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GENERATED DATASET

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Records: 1000 | Parameters: 10

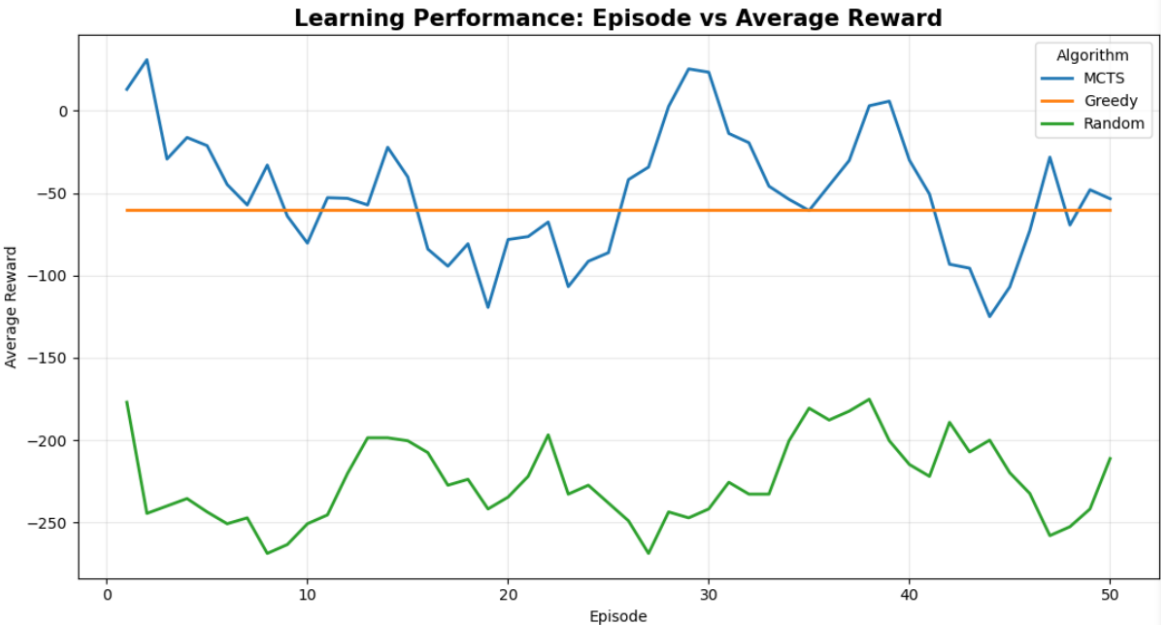
isode	State_X	State_Y	Goal_X	Goal_Y	Distance_to_Goal	Obstacle_Density	Step_Cost	Goal_Reward	Collision_Penalty
1	0	6	7	7	8	0.140625	-1	100	-10
2	5	3	7	7	6	0.140625	-1	100	-10
3	3	6	7	7	5	0.140625	-1	100	-10
4	0	5	7	7	9	0.140625	-1	100	-10
5	1	0	7	7	13	0.140625	-1	100	-10
6	4	7	7	7	3	0.140625	-1	100	-10
7	5	6	7	7	3	0.140625	-1	100	-10
8	5	6	7	7	3	0.140625	-1	100	-10
9	4	1	7	7	9	0.140625	-1	100	-10
10	6	3	7	7	5	0.140625	-1	100	-10

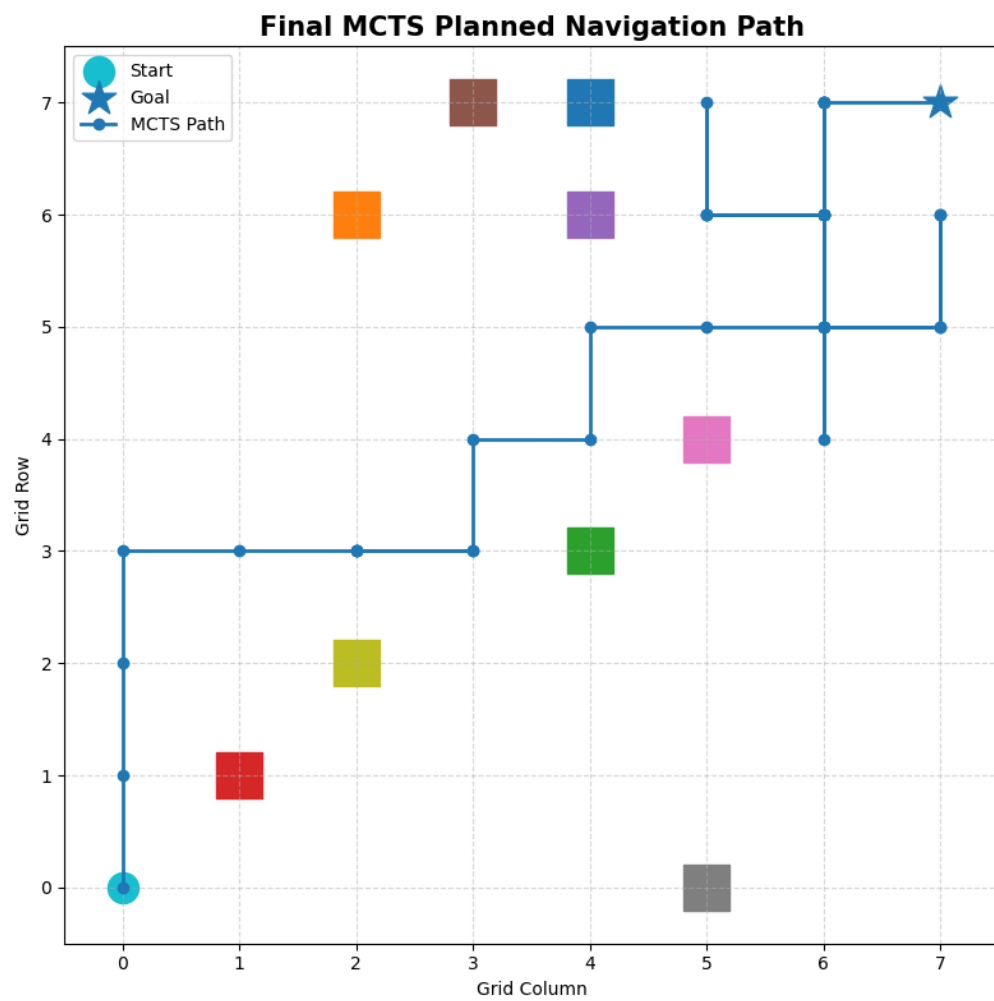
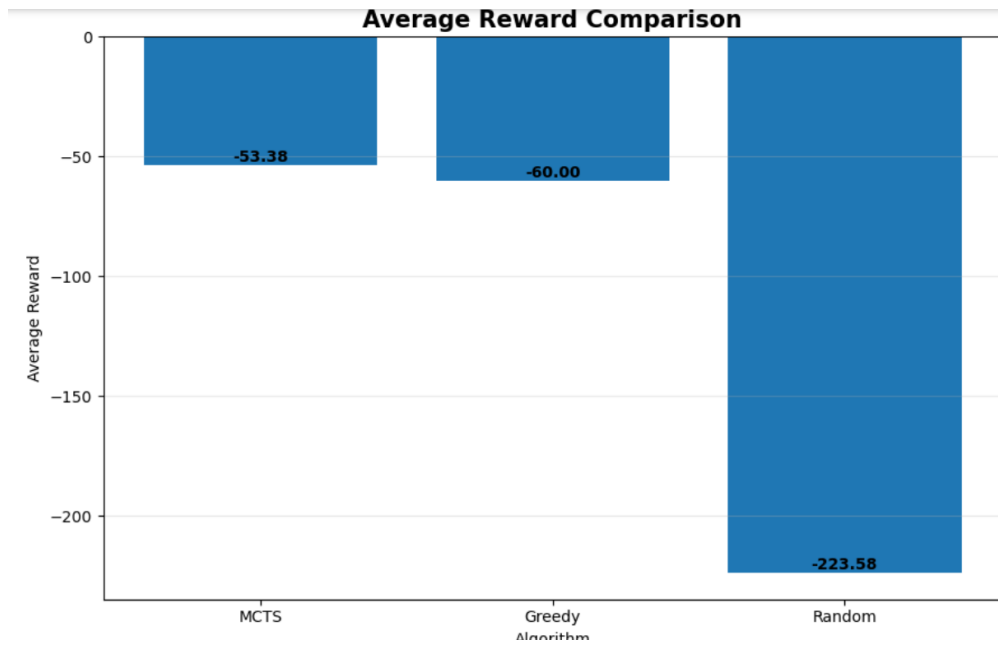
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FINAL PERFORMANCE SUMMARY

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	Algorithm	Average_Reward	Reward_Std	Success_Rate	Average_Steps	Average_Collisions
0	Greedy	-60.00	0.00	0.0	60.0	0.00
1	MCTS	-53.38	83.60	66.0	44.8	8.36
2	Random	-223.58	61.85	2.0	60.0	18.40





Result:

The sampling-based planning model was successfully designed and implemented using **Monte Carlo Tree Search (MCTS)**. A synthetic grid-world dataset was generated and the MCTS planner was evaluated against Greedy and Random baseline approaches. The models were compared using reward, success rate, navigation steps, and collision metrics. The generated graphs and summary table provide a clear evaluation of the planning performance, with the best-performing approach identified from the experimental results.

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FINAL EVALUATION
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Best Algorithm      : MCTS
Average Reward      : -53.38
Success Rate        : 66.00%
Average Steps        : 44.80
Average Collisions   : 8.36

Final MCTS Episode
Reward              : 37
Steps               : 37
Success             : 1
Collisions          : 3

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IMPLEMENTATION COMPLETED SUCCESSFULLY
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